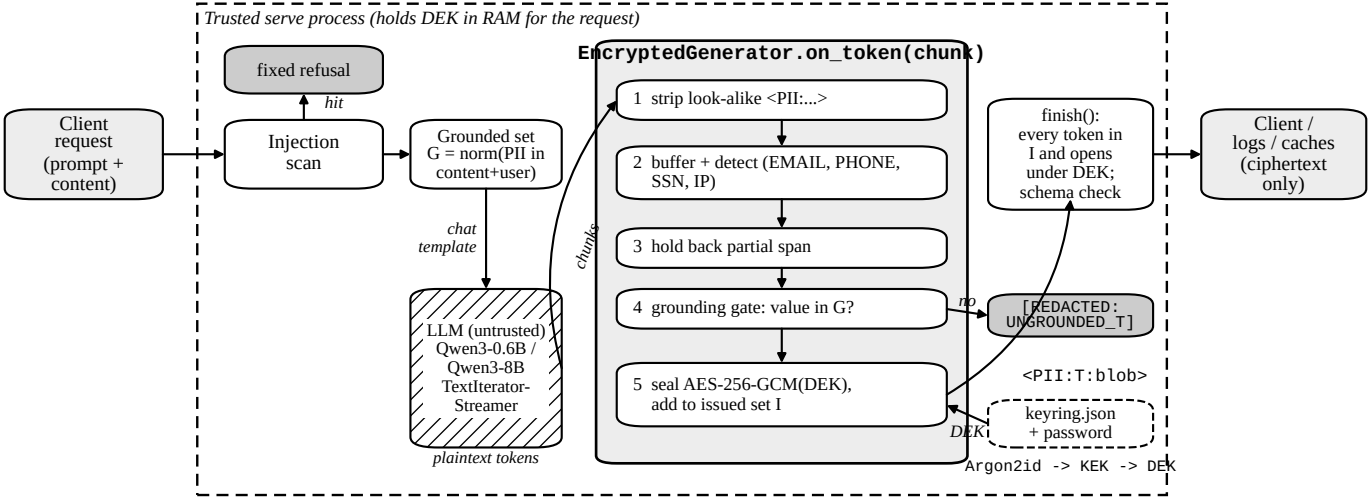
Trusted serve process (holds DEK in RAM for the request)
Client request (prompt + content)
Injection scan
hit
fixed refusal
Grounded set G = norm(PII in content+user)
chat template
LLM (untrusted) Qwen3-0.6B / Qwen3-8B TextIterator-Streamer
plaintext tokens
chunks
EncryptedGenerator.on_token(chunk)
1 strip look-alike <PII:...>
2 buffer + detect (EMAIL, PHONE, SSN, IP)
3 hold back partial span
4 grounding gate: value in G?
no
[REDACTED: UNGROUNDED_T]
5 seal AES-256-GCM(DEK), add to issued set I
<PII:T:blob>
DEK
keyring.json + password
Argon2id -> KEK -> DEK
finish(): every token in I and opens under DEK; schema check
Client / logs / caches (ciphertext only)